

PORTFOLIO CASE STUDY

Logistics Performance Analysis

Revenue, customer mix, lane economics, and delivery reliability

A detailed business intelligence report built from a Kaggle-derived logistics dataset

Prepared by	Analysis stack	Reporting scope
Andrew Boadi	MySQL • Tableau Public	2022–2024 complete years

Headline conclusion

Revenue remained broadly stable while known costs declined, but fewer than half of deliveries were on time. The strongest opportunity is therefore operational: improve network reliability while protecting high-value customers and lanes.

Source note: The dataset was derived from Kaggle. The original dataset URL was not supplied in the project files.

Document Guide

This report is the long-form companion to the repository README. It explains the business questions, SQL methodology, quality controls, Tableau design decisions, findings, and recommended actions in enough detail for a hiring manager or analyst to evaluate the work.

Section	Purpose
1. Executive Summary	Condensed business story and priority actions
2. Scope & Questions	Coverage, exclusions, and analytical objectives
3. Data & Methodology	Tables, relationships, cleaning, validation, and SQL approach
4. Findings	Financial, customer, lane, and delivery insights
5. Dashboard Review	Four portfolio dashboards and how to read them
6. Recommendations	Prioritized business actions linked to evidence
7. Definitions & Limitations	Metric logic, caveats, and future work
Appendix	Repository structure and representative SQL patterns

How to read the report

- Blue and teal are used for neutral or positive analytical emphasis.
- Red and orange indicate delivery reliability risk, not relative success.
- Grey states on the delivery map mean no data / not served—not 0% on-time performance.
- All dollar amounts shown with M are millions; rates use decimal points throughout.

1. Executive Summary

The analysis evaluates a logistics operator across four connected dimensions: financial performance, customer contribution, lane economics, and delivery reliability. MySQL was used to validate relationships, clean and aggregate the operational data, and create Tableau-ready outputs. Tableau Public was used to build four dashboards that translate the SQL results into a business narrative.

Headline KPI	Result	Interpretation
2024 gross revenue	\$99.79M	Recovered close to the 2022 level after a modest 2023 decline
Known operating costs	\$37.30M → \$32.75M	Available fuel, maintenance, and claims costs fell by \$4.55M
Estimated known-cost margin	62.67% → 67.19%	Improved because recorded costs declined; not a full operating margin
Largest customer segment	Contract: \$112.33M	Contract revenue led, but Spot and Dedicated were also material
Top gross-revenue lane	NC → OR: \$11.23M	Scale leader
Top revenue-per-mile lane	PA → NY: \$3.62/mile	Rate leader, but low total contribution
Overall on-time rate	44.6%	Fewer than half of deliveries arrived on time
Late deliveries	47,264	Most urgent operational issue

Executive interpretation

- The company was stable, not meaningfully growing: annual revenue stayed near \$99M across 2022–2024.
- Recorded margin improved through lower known costs, although the dataset omits several major expense categories.
- Lane leaders differed depending on the metric. Total contribution, revenue per mile, revenue per load, distance, and volume must be assessed together.
- Delivery underperformance was network-wide: every served state remained below 46% on-time.

Management priority

Treat service reliability as a cross-functional transformation program, not a local exception. Protect the revenue base while diagnosing delays by state, facility, customer, lane, and delay reason.

2. Business Context, Scope & Questions

Project objective

The objective was to convert a multi-table logistics dataset into a decision-oriented portfolio project. The analysis asks not only what happened, but how customer mix, route economics, operational costs, and delivery performance fit together.

Reporting scope

Item	Treatment
Source	Kaggle-derived logistics dataset; exact source URL not supplied
Broad source coverage	2020 to early 2025
Primary analytical period	Complete business years 2022, 2023, and 2024
Excluded years	2020 and 2021 were excluded from the final story; partial 2025 was excluded from annual comparisons
Tools	MySQL for analysis and preparation; Tableau Public for visualization
Reusable outputs	Analyst-created CSV extracts / view outputs for dashboard development

Business questions

- Was revenue growing, declining, or broadly flat?
- Did the cost base improve, and what can the available data support about margin?
- Which customers and customer types generated the most revenue?
- Which lanes led by total revenue and by rate efficiency—and why were they different?
- Was poor delivery reliability concentrated in a few states or systemic across the network?

3. Data Architecture & Analytical Workflow

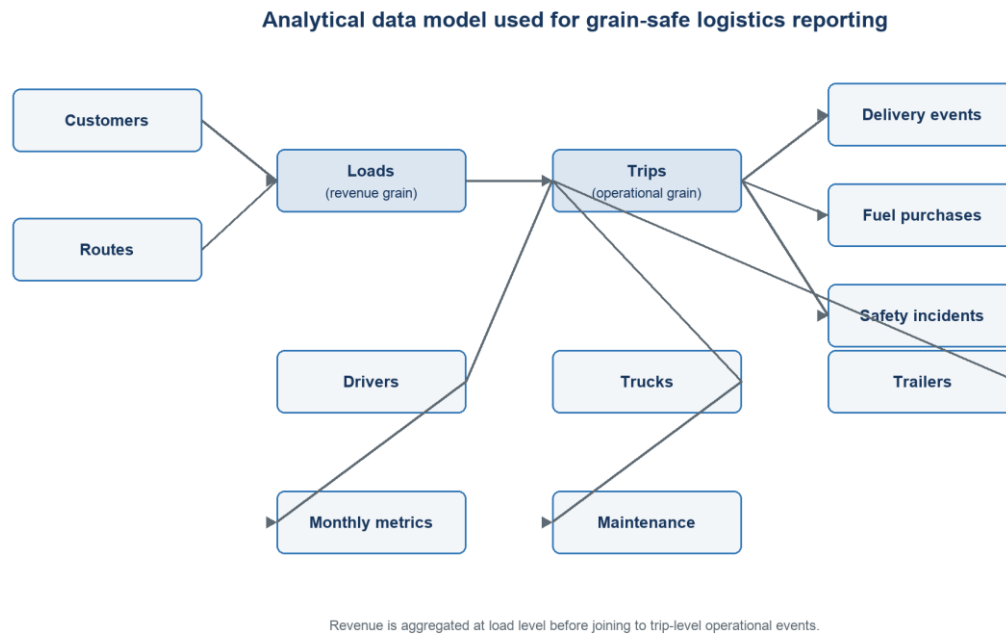
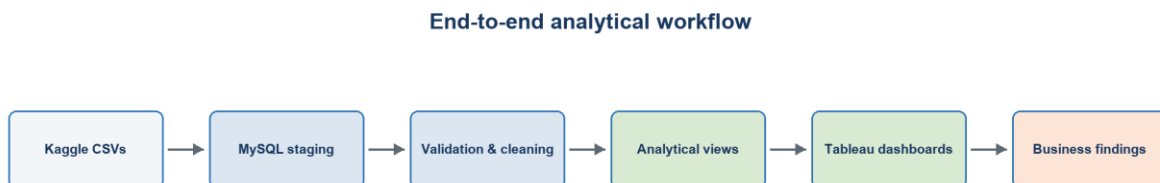


Figure 1. Simplified analytical data model and grain-safe join strategy.

Core modeling principle: preserve grain

Revenue belongs to loads, while many operational measures belong to trips or events. A direct one-to-many join can duplicate revenue. The SQL therefore aggregates each source to the required reporting grain before combining results. This was one of the most important quality controls in the project.



Raw operational data was transformed into reusable Tableau-ready outputs and an evidence-based business narrative.

Figure 2. End-to-end workflow from raw CSV files to business findings.

4. Dataset Inventory

The repository contains 14 CSV files and 549,706 total rows. The tables cover commercial demand, asset utilization, trips, delivery events, fuel, maintenance, safety, and facility data.

File	Rows	Analytical role
customers.csv	200	Customer identity, type, and commercial attributes
loads.csv	85,410	Shipment revenue and load attributes
trips.csv	85,410	Operational movement connecting loads, routes, assets, and drivers
delivery_events.csv	170,820	Pickup/delivery timing events
fuel_purchases.csv	196,442	Fuel spend and purchase activity
maintenance_records.csv	2,920	Truck maintenance events and costs
safety_incidents.csv	170	Safety events and claims costs
driver_monthly_metrics.csv	4,464	Driver performance by month
truck_utilization_metrics.csv	3,312	Truck utilization by month
routes.csv	58	Origin/destination route definition
drivers.csv	150	Driver master data
trucks.csv	120	Truck master data
trailers.csv	180	Trailer master data
facilities.csv	50	Facility and location master data

Coverage note

Loads and trips span 2022–2024. Delivery and fuel event files extend into early January 2025, but the partial 2025 period was excluded from annual trend comparisons.

5. Data Cleaning, Validation & Preparation

The project SQL is both an analysis script and an audit trail. It contains relationship checks, business summaries, trend analysis, customer analysis, and geographic/lane analysis. The following controls were applied before results were presented in Tableau.

Control	Implementation / rationale
Relationship validation	Checked whether a load connected to multiple trips before combining load revenue with operational tables.
Date validation	Reviewed table-specific date ranges and restricted annual storytelling to complete 2022–2024 periods.
Null and denominator safety	Used NULLIF in rate calculations to avoid divide-by-zero errors.
Revenue definition	Standardized gross revenue as linehaul revenue + fuel surcharge + accessorial charges.
Grain-safe aggregation	Summarized load revenue, trip activity, and cost/event measures independently before joining.
Duplicate-risk control	Avoided summing load revenue after one-to-many joins that could multiply financial values.
Ranking logic	Used ROW_NUMBER and ordered aggregates for top-customer and within-year rankings.
Trend consistency	Used common year fields and consistent aggregation to compare revenue, costs, and estimated margin.
Presentation standardization	Formatted millions with decimal points and used consistent percent notation.
Map semantics	Separated served states with data from grey states representing no data / not served.

Representative SQL techniques

- Common table expressions (CTEs)
- Conditional aggregation
- Window functions including ROW_NUMBER
- LAG for year-over-year comparison
- CASE expressions for tiering and segmentation
- COALESCE and NULLIF for robust metrics
- Explicit grain-specific aggregation before joins

Reproducibility note

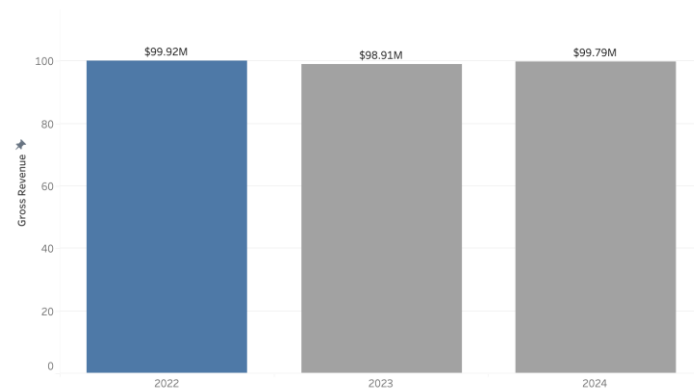
The repository includes the analytical SQL used to calculate the reported metrics. Tableau source CSVs were created by the analyst, but the current SQL file does not contain every final CREATE VIEW statement; they are therefore documented as analytical outputs rather than claimed as fully reproducible database objects.

6. Finding 1 — Stable Revenue, Lower Known Costs

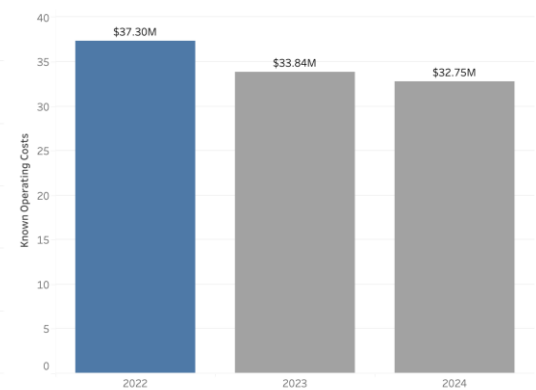
Logistics Business Performance Overview

Gross Revenue Stayed broadly flat from 2022 to 2024 while known operating costs declined and estimated margin improved

Gross Revenue by Year



Known Operating Cost by Year



Estimated Margin Improved by Year

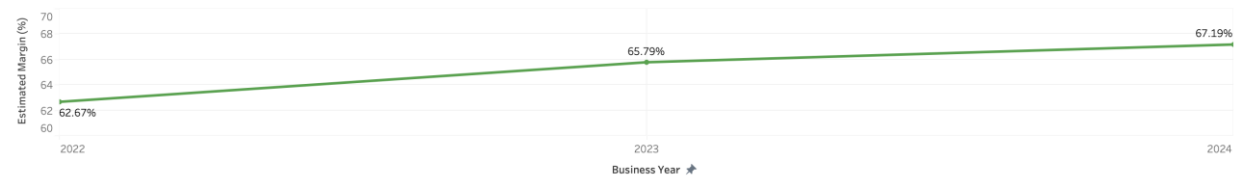


Figure 3. Logistics Business Performance Overview dashboard.

Year	Gross revenue	Known costs	Estimated known-cost margin
2022	\$99.92M	\$37.30M	62.67%
2023	\$98.91M	\$33.84M	65.79%
2024	\$99.79M	\$32.75M	67.19%

Interpretation

Revenue was broadly flat: 2024 recovered close to the 2022 level after a small decline in 2023. The stronger movement was in recorded operating costs, which declined by \$4.55M from 2022 to 2024. That mechanically improved the estimated margin based on the cost categories available in the data.

Important limitation

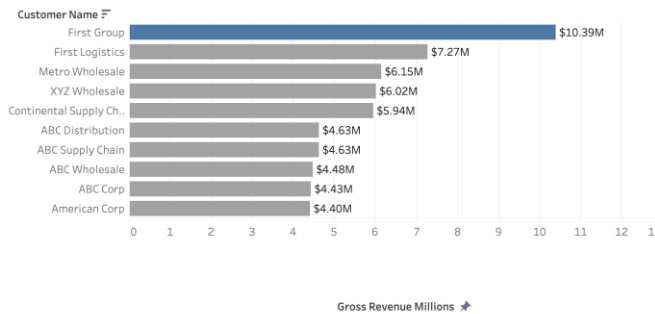
Known operating costs include fuel, maintenance, and claims. They do not include wages, insurance, rent, depreciation, taxes, or administrative overhead; the estimated margin must not be presented as a full GAAP or operating profit margin.

7. Finding 2 — Customer Revenue Was Broadly Distributed

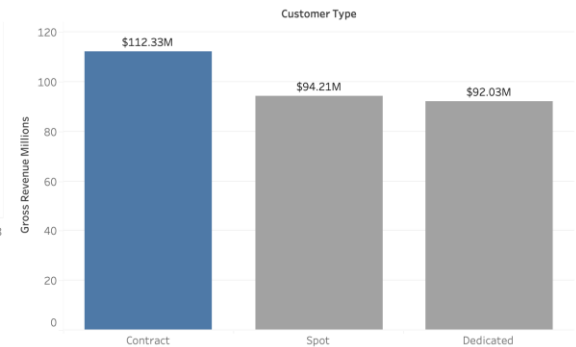
Customer Revenue & Freight Mix Overview

Revenue is led by Contract customers, while freight mix shows which types of work drive each customer segment

Top 10 Customers by Gross Revenue



Revenue By Customer Type



Revenue Mix by Customer Type and Freight Type

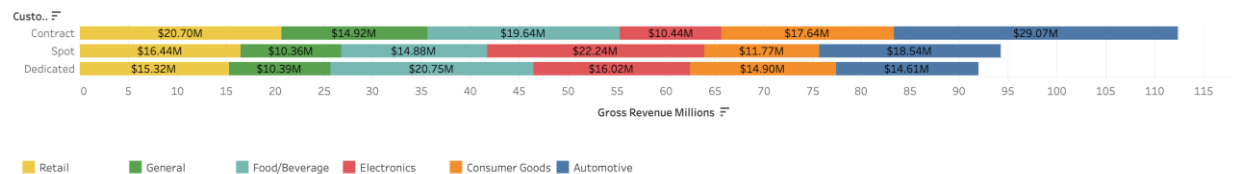


Figure 4. Customer Revenue & Freight Mix Overview dashboard.

Customer segment	Gross revenue	Interpretation
Contract	\$112.33M	Largest segment
Spot	\$94.21M	Material secondary segment
Dedicated	\$92.03M	Close to Spot and strategically important

First Group was the largest individual customer at \$10.39M. It was important but did not dominate the company's nearly \$299M three-year revenue base. This suggests the portfolio was reasonably distributed at the individual account level, while customer-type mix remained balanced enough to avoid dependence on a single commercial model.

Management implications

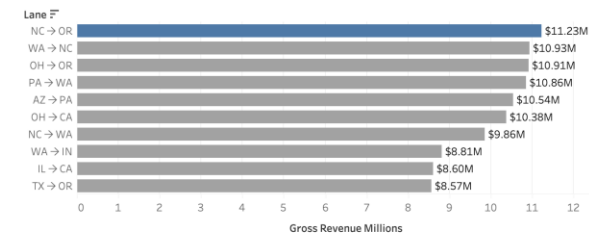
- Protect top accounts with proactive service reviews and retention plans.
- Track customer concentration over time, not only at one point in aggregate.
- Use freight-type mix to identify cross-sell opportunities and operational dependencies within each segment.

8. Finding 3 — Revenue Scale and Rate Efficiency Are Different

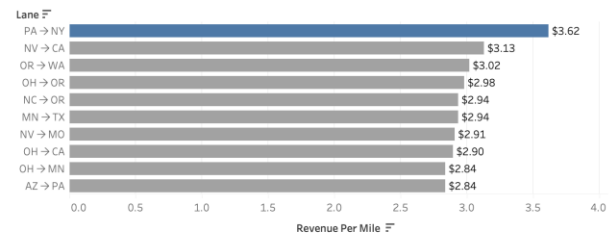
Lane & Route Performance

Revenue leaders are not always rate-efficiency leaders—haul distance and revenue per load help explain the difference.

Top 10 Lanes by Gross Revenue



Top 10 Lanes by Revenue per Mile



Rate Efficiency vs Revenue Scale: Bubble size represents average gross revenue per load

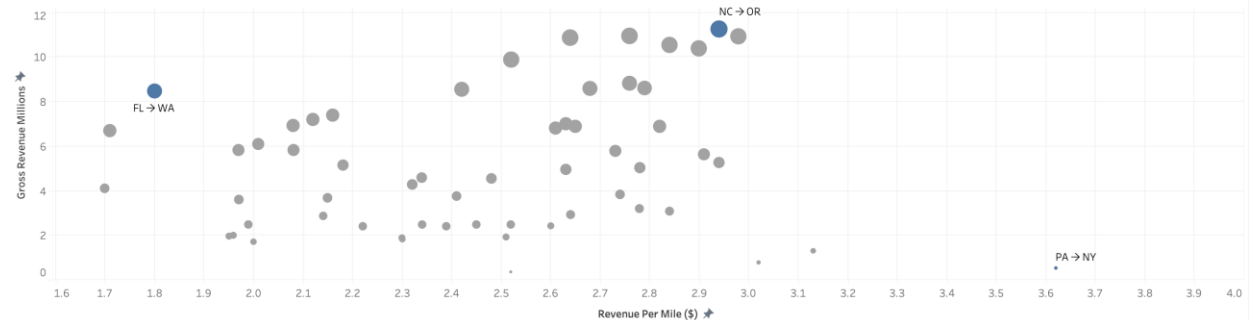


Figure 5. Lane & Route Performance dashboard.

The top bar charts answer two different questions: which lanes generate the most revenue, and which lanes earn the most per mile. The scatterplot combines rate and total revenue, while bubble size represents average gross revenue per load. Reading all three encodings together prevents a high rate from being mistaken for high total value.

Lane	Gross revenue	Revenue / mile	Loads	Miles / load	Revenue / load
NC → OR	\$11.23M	\$2.94	1,410	2,708	\$7,964
PA → NY	~\$0.52M	\$3.62	~1,537	~94	~\$341
FL → WA	~\$8.45M	\$1.80	~1,450	~3,230	~\$5,825

PA → NY versus FL → WA

PA → NY was the clearest high-rate, low-scale example. Its short average haul produced the highest revenue per mile, but low revenue per load limited total contribution. FL → WA showed the opposite profile: a lower rate per mile but much higher total revenue because each shipment covered a long distance and generated far more revenue per load.

Decision rule

Evaluate lanes with a balanced scorecard: volume, revenue per mile, revenue per load, haul distance, service reliability, and cost-to-serve. No single rate metric is sufficient.

9. Finding 4 — Delivery Reliability Is the Critical Risk

Delivery Reliability Overview

On-time delivery remains below 46% in every state served, indicating a system-wide reliability problem

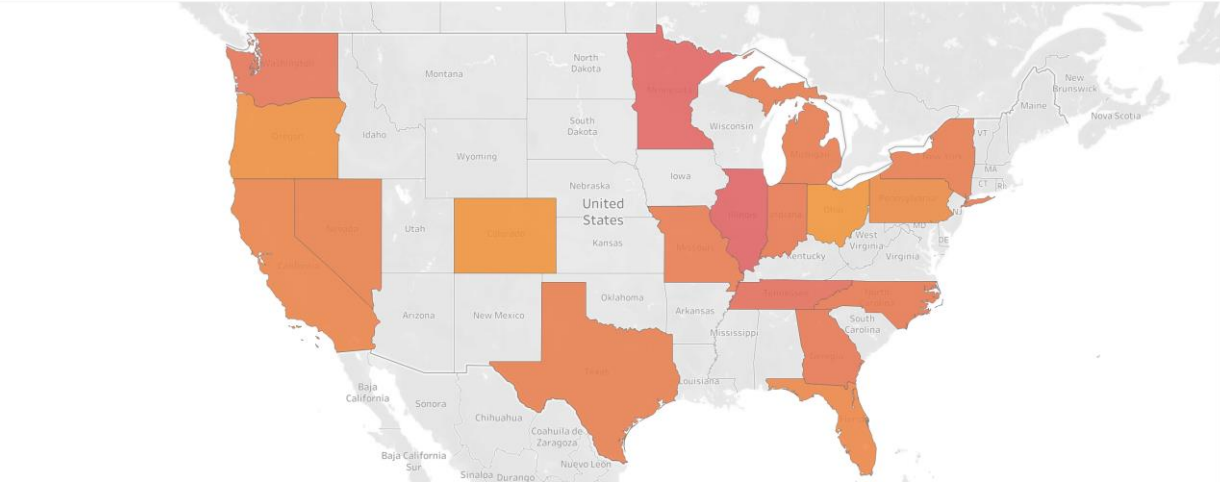


Figure 6. Delivery Reliability Overview dashboard. Grey states indicate no delivery data / not served.

Reliability KPI	Result	Meaning
Overall on-time rate	44.6%	Fewer than half of deliveries were on time
Total deliveries	85,333	Observed deliveries in the KPI scope
Late deliveries	47,264	55.4% of deliveries were late
State range	42.9%–45.9%	No served state achieved an acceptable result

The narrow state-level range matters: poor performance was not confined to one outlier market. The evidence points to a network-wide reliability issue that may involve planning, dispatch, facility operations, asset availability, appointment management, or data-quality rules used to classify delivery timing.

Interpretation guardrail

Grey states are not 0% performers. They have no delivery data in this reporting view or were not served. They must remain visually distinct from the red–orange performance scale.

10. Prioritized Recommendations

#	Action	Evidence	Next analytical step
1	Launch a network-wide late-delivery root-cause review	44.6% on time across every served state	Segment delays by reason, facility, customer, lane, day, dispatcher, asset, and appointment window
2	Prioritize high-volume sources of lateness	47,264 late deliveries	Rank by absolute late count and late rate; address the highest operational burden first
3	Create service-recovery KPIs	State averages alone hide process detail	Track appointment adherence, dwell time, tender lead time, missed windows, and delay reasons
4	Protect top revenue lanes	NC → OR and peers materially support revenue	Monitor service, volume stability, concentration, and cost-to-serve
5	Test growth on high-rate, low-scale lanes	PA → NY has the strongest unit rate	Assess demand, capacity, customer fit, and minimum profitable volume before expanding
6	Evaluate long-haul economics beyond revenue	FL → WA shows scale but low RPM	Add fuel, driver, equipment, repositioning, and empty-mile economics
7	Continue cost control cautiously	Known costs fell \$4.55M	Validate missing expense categories before making profitability claims
8	Monitor customer retention and concentration	First Group leads; Contract is largest segment	Add account trend, churn risk, service performance, and margin by customer

Suggested operating cadence

- Weekly: late-delivery exception review for highest-volume states, facilities, and lanes.
- Monthly: customer and lane scorecards combining revenue, service, rate, and cost-to-serve.
- Quarterly: financial review reconciling known-cost margin with a more complete operating expense model.

11. Dashboard Design Rationale

Dashboard	Visual structure	Why it works
Business Performance	Annual bars + margin line	Separates scale from direction and highlights stable revenue versus falling costs
Customer Revenue & Freight Mix	Ranked bars + stacked mix	Shows individual account leaders, segment scale, and work mix
Lane & Route Performance	Dual top-10 bars + bubble scatter	Contrasts total contribution with rate efficiency and load economics
Delivery Reliability	KPI strip + filled map	Makes the system-wide service risk immediately visible

Visual conventions

- One accent color is used to highlight the leading bar or selected lane; the remaining marks are grey to reduce noise.
- The delivery dashboard uses red-to-orange because even the best state remains objectively weak; green would imply acceptable performance.
- Labels are limited to leaders and meaningful outliers to reduce scatterplot clutter.
- Decimal points are used consistently: \$11.23M, \$3.62/mile, and 62.67%.
- Tooltips carry supporting metrics such as total deliveries, late deliveries, and lane economics without overcrowding the view.

Portfolio storytelling sequence

1. Begin with the overall financial position.
2. Explain who generates the revenue.
3. Show where revenue and unit-rate efficiency diverge.
4. Close with the most actionable operational risk: delivery reliability.

12. Metric Definitions

Metric	Definition
Gross revenue	Linehaul revenue + fuel surcharge + accessorial charges
Known operating costs	Fuel purchase cost + maintenance cost + safety/claims cost available in the dataset
Estimated known-cost margin	$(\text{Gross revenue} - \text{known operating costs}) \div \text{gross revenue}$
Revenue per mile	$\text{Gross revenue} \div \text{actual miles}$
Revenue per hour	$\text{Gross revenue} \div \text{actual trip duration hours}$
Average gross revenue per load	$\text{Gross revenue} \div \text{total loads}$
On-time delivery rate	$\text{On-time deliveries} \div \text{total deliveries}$
Late deliveries	Deliveries classified as not on time under the delivery-event logic

Known limitations

- The dataset is synthetic or portfolio-oriented and does not represent a complete audited operating system.
- Known costs do not include all labor, insurance, rent, tax, depreciation, financing, or overhead expenses.
- The original Kaggle URL and license details were not included in the local project files.
- The annual narrative excludes 2020–2021 and partial 2025, so it should not be generalized to those periods.
- On-time performance is only as reliable as the event timestamps and classification rule in the source data.
- Grey states represent no data / not served, not zero performance.
- The current SQL script documents analytical queries but not every final Tableau extract creation statement.

Appendix A — Repository & Reproduction Guide

Path	Contents
README.md	Portfolio-facing project narrative
sql/logistics_analysis.sql	MySQL analytical queries and commentary
DATABASE_SCHEMA.txt	Source table structure and relationships
images/	Four exported Tableau dashboard screenshots
reports/	This detailed Word case-study report
*.csv	Fourteen source datasets

Suggested reproduction steps

5. Create a MySQL database and load the 14 CSV files using the schema as a reference.
6. Run relationship and date-range checks before joining tables.
7. Execute the analytical SQL sections for business, trend, customer, and route analysis.
8. Export the required query/view results as Tableau-ready CSV files.
9. Connect Tableau Public to the prepared extracts and rebuild the worksheets and dashboards.
10. Validate dashboard totals against SQL outputs before publishing.

Representative grain-safe SQL pattern

```

WITH load_summary AS (
  SELECT load_id,
         linehaul_revenue + fuel_surcharge + accessorial_charges AS gross_revenue
  FROM loads
),
trip_summary AS (
  SELECT load_id, SUM(actual_miles) AS actual_miles
  FROM trips
  GROUP BY load_id
)
SELECT SUM(l.gross_revenue) AS gross_revenue,
       SUM(t.actual_miles) AS actual_miles
FROM load_summary l
LEFT JOIN trip_summary t USING (load_id);

```

Pattern shown for illustration; consult sql/logistics_analysis.sql for the project's complete analytical queries.

Appendix B — Further Analysis Opportunities

The current project answers the core portfolio questions. A production-grade extension could deepen the operational diagnosis and make the dashboards more actionable.

Extension	Recommended measures
Reliability drivers	Delay reason, facility dwell, appointment windows, dispatch lead time, driver/asset availability
True lane profitability	Fuel, labor, maintenance allocation, tolls, repositioning, empty miles, and accessorial recovery
Customer value	Gross margin, service cost, claims exposure, retention, and growth trend by customer
Forecasting	Monthly volume, revenue, fuel expense, and capacity forecasts
Operational benchmarking	Targets by state, facility, lane, customer, and freight type
Data engineering	Version-controlled CREATE VIEW scripts and automated validation tests

Closing statement

This project demonstrates end-to-end analytical thinking: understanding relational grain, validating and preparing data in MySQL, defining transparent metrics, building coherent Tableau dashboards, and translating results into business decisions. The clearest conclusion is that cost discipline improved the available financial picture, but sustainable performance depends on repairing delivery reliability without weakening the customer and lane revenue base.

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